

Sail power

Early sailing ships mostly had square sails, which worked only with the wind pushing from behind. Then, 2,000 years ago, the Romans invented triangular sails, which came to be called Latin or lateen sails. These can be set at an angle, allowing a ship to sail almost directly into the wind.

THREE-MASTERS

From the 1500s, big sailing ships with two, three, and even four masts sailed the oceans. The huge sail area caught enough wind for the vessels to carry heavy cargo. The first battleships were warships, known as “men o’ war,” which could even carry many heavy cannons.



**Model of 1660
Dutch man o' war**

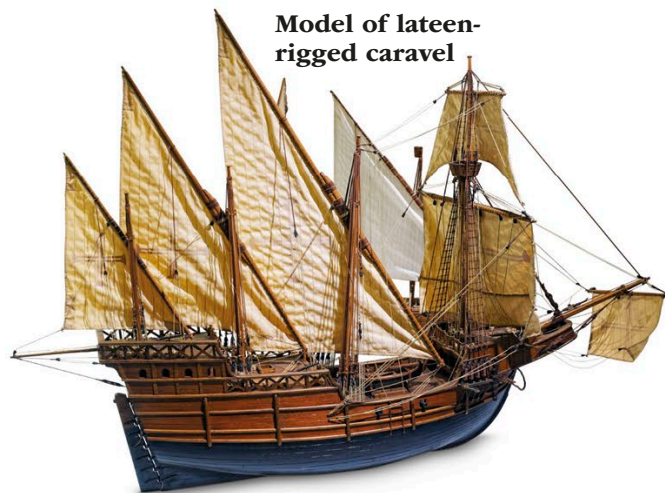
*Cannons
mounted
on the ship*



LIFE JACKET

The forerunner of modern life jackets was invented by the British Arctic explorer Captain John Ward in 1854. Made from blocks of cork, it was uncomfortable to wear. Today life jackets use soft foam-filled bags, or inflate with gas from a carbon-dioxide cylinder when needed.

**Model of lateen-
rigged caravel**



THE CARAVEL

Sturdy, square-sailed ships were fine for coastal waters, but not for oceans. In the 1400s, European sailors began to use fast, light vessels called “caravels,” which had lateen sails. In the caravels, sailors could travel across the ocean, knowing they could sail long distances quickly, and get home whatever the wind direction.

► THE FLYING YACHT

The Oracle team's yacht for the America's Cup race is designed almost to fly over the water at speeds of 60 mph (96.5 km/h). Hydrofoils, or “daggerboards,” help lift the yacht to glide above the water, reducing friction.



*Streamlined twin
bulls made of tough,
light carbon fiber*